Exercise 11.3

SOLUTION

Q1. Find the maximum and minimum values of the following functions.

i. $3-\sin 3x$

Sol: $f(x)=3-\sin 3x$

Here $a=3, \quad b=-1$

Maximum value of $f(x)=a+|b|=3+|(-1)|=3+1=4$

Minimum value of $f(x)=a-|b|=3-|(-1)|=3-1=2$

ii. $3+\sin 2x$

Sol: $f(x)=3+\sin 2x$

Here $a=3, \quad b=1$

Maximum value of $f(x)=a+|b|=3+|1|=3+1=4$

Minimum value of $f(x)=a-|b|=3-|1|=3-1=2$

iii. $\frac{1}{2}+\sin(5x+\pi)$

Sol: $f(x)=\frac{1}{2}+\sin(5x+\pi)$

Here $a=\frac{1}{2}, \quad b=1$

Maximum value of $f(x)=a+|b|=\frac{1}{2}+|1|=\frac{1}{2}+1=\frac{3}{2}$

Minimum value of $f(x)=a-|b|=\frac{1}{2}-|1|=\frac{1}{2}-1=-\frac{1}{2}$

iv. $\frac{3}{2}+\cos\left(x-\frac{\pi}{4}\right)$

Sol: $f(x)=\frac{3}{2}+\cos\left(x-\frac{\pi}{4}\right)$

Here $a=\frac{3}{2}, \quad b=1$

Maximum value of $f(x)=a+|b|=\frac{3}{2}+|1|=\frac{3}{2}+1=\frac{5}{2}$

Minimum value of $f(x)=a-|b|=\frac{3}{2}-|1|=\frac{3}{2}-1=\frac{1}{2}$

v. $1-3\cos 2x$

Sol: $f(x)=1-3\cos 2x$

Here $a=1, \quad b=-3$

Maximum value of $f(x)=a+|b|=1+|-3|=1+3=4$

Minimum value of $f(x)=a-|b|=1-|-3|=1-3=-2$

vi. $1+2\sin\left(x+\frac{\pi}{6}\right)$

Sol: $f(x)=1+2\sin\left(x+\frac{\pi}{6}\right)$

Here $a=1, \quad b=2$

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Maximum value of $f(x) = a + |b| = 1 + |2| = 1 + 2 = 3$

Minimum value of $f(x) = a - |b| = 1 - |2| = 1 - 2 = -1$

vii. $\dfrac{1}{10 - 2\sin 3x}$

Sol: $g(x) = \dfrac{1}{10 - 2\sin 3x}$

Here $a = 10,\ b = -2$

Maximum value of $g(x) = \dfrac{1}{a - |b|} = \dfrac{1}{10 - |-2|} = \dfrac{1}{10 - 2} = \dfrac{1}{8}$

Minimum value of $g(x) = \dfrac{1}{a + |b|} = \dfrac{1}{10 + |-2|} = \dfrac{1}{10 + 2} = \dfrac{1}{12}$

viii. $\dfrac{1}{7 + 3\cos(-2x)}$

Sol: $g(x) = \dfrac{1}{7 + 3\cos(-2x)}$

Here $a = 7,\ b = 3$

Maximum value of $g(x) = \dfrac{1}{a - |b|} = \dfrac{1}{7 - |3|} = \dfrac{1}{7 - 3} = \dfrac{1}{4}$

Minimum value of $g(x) = \dfrac{1}{a + |b|} = \dfrac{1}{7 + |3|} = \dfrac{1}{7 + 3} = \dfrac{1}{10}$

ix. $\dfrac{1}{5 - 3\cos(3x - 1)}$

Sol. $g(x) = \dfrac{1}{5 - 3\cos(3x - 1)}$

Here $a = 5,\ b = -3$

Maximum value of $g(x) = \dfrac{1}{a - |b|} = \dfrac{1}{5 - |-3|} = \dfrac{1}{5 - 3} = \dfrac{1}{2}$

Minimum value of $g(x) = \dfrac{1}{a + |b|} = \dfrac{1}{5 + |-3|} = \dfrac{1}{5 + 3} = \dfrac{1}{8}$

Q2. **The temperature T in degrees Celsius of a certain city varies throughout the day according to the equation $T(t) = \dfrac{13}{2}\sin\left(\dfrac{\pi}{6}t - \dfrac{\pi}{9}\right) + 15$, where t is the time in hours, with t = 0 corresponding to midnight.**

Sol. $T(t) = \dfrac{13}{2}\sin\left(\dfrac{\pi}{6}t - \dfrac{\pi}{9}\right) + 15$

Here $a = 15,\ b = \dfrac{13}{2}$

a) **Find the Maximum and Minimum temperature during the day**

Maximum temperature $= a + |b| = 15 + \left|\dfrac{13}{2}\right|$

$= 15 + \dfrac{13}{2} = 15 + 6.5 = 21.5^{\circ}$

Minimum temperature $= a - |b| = 15 - \left|\dfrac{13}{2}\right|$

$= 15 - 6.5 = 8.5^{\circ}$

b) **Find the Temperature at t = 9hours (9:00 a.m.)**

$T(9) = \dfrac{13}{2}\sin\left(\dfrac{\pi}{6}(9) - \dfrac{\pi}{9}\right) + 15$

$= \dfrac{13}{2}\sin\left(\dfrac{25\pi}{18}\right) + 15 = 8.89^{\circ}$

Q.3 **A man on the top of a 100 m high light-house is in line with two ships on the same side of it, whose angles of depression from the man are 17° and 19° respectively. Find the distance between the ships.**

Sol: Let d_1 be the distance from the lighthouse to the farther ship.

$\tan(17^{\circ}) = \dfrac{100}{d_1}$

$d_1 = \dfrac{100}{\tan(17^{\circ})}$

$d_1 \approx 327.10m$

Let d_2 be the distance from the lighthouse to the closer ship.

$\tan(19^{\circ}) = \dfrac{100}{d_2}$

$d_2 = \dfrac{100}{\tan(19^{\circ})}$

$d_2 \approx 290.42m$

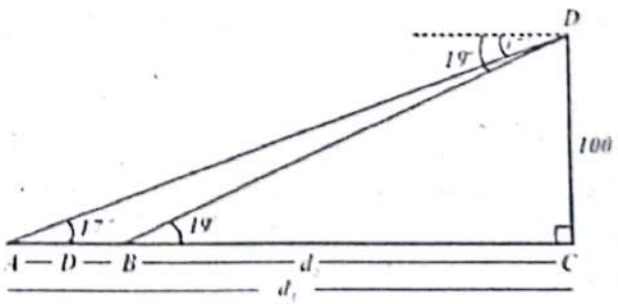
Let D be the distance between the ships.

$D = d_1 - d_2$

$D = 327.10 - 290.42$

$D \approx 36.68m$

The distance between the two ships is approximately $36.68m$.



Q.4 **P and Q are two points in line with a tree. If the distance between P and Q be $30m$ and the angles of elevation of the top of the tree at P and Q be 12° and 15° respectively, find the height of the tree.**

Sol: Let h be the height of the tree.
Let x be the distance from point Q to the base of the tree.

$$\tan(15^\circ)=\frac{h}{x}$$

$$\tan(12^\circ)=\frac{h}{x+30}$$

$$x=\frac{h}{\tan(15^\circ)}$$

$$\tan(12^\circ)=\frac{h}{x+30}=\frac{h}{\frac{h}{\tan(15^\circ)}+30}$$

$$\tan(12^\circ)\left(\frac{h}{\tan(15^\circ)}+30\right)=h$$

$$\frac{h\tan(12^\circ)}{\tan(15^\circ)}+30\tan(12^\circ)=h$$

$$h-\frac{h\tan(12^\circ)}{\tan(15^\circ)}=30\tan(12^\circ)$$

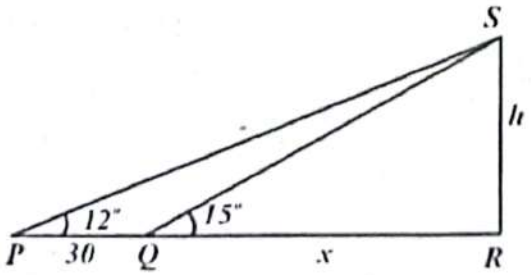
$$h\left(1-\frac{\tan(12^\circ)}{\tan(15^\circ)}\right)=30\tan(12^\circ)$$

$$h=\frac{30\tan(12^\circ)}{1-\frac{\tan(12^\circ)}{\tan(15^\circ)}}=\frac{30\tan(12^\circ)\tan(15^\circ)}{\tan(15^\circ)-\tan(12^\circ)}$$

$$h=\frac{30\times0.2126\times0.2679}{0.2679-0.2126}=\frac{1.705}{0.0553}$$

$$h\approx30.83m$$

The height of the tree is approximately $30.83m$



Q.5 **A gaint Ferris wheel has a diameter of 60feet. The lowest point of the wheel is located 6 feet above the ground. The wheel completes on full revolution every 80 seconds.**

(a) **Model an equation that represent the height h(t) of rider on the Ferris wheel at any given time t.**

(b) **Find the maximum height of a rider**

(c) **Find the height of the rider from the ground after 35 seconds.**

Sol: $Radius=r=\frac{60}{2}=30\ feet$

To find the angular frequency B:

$$B=\frac{2\pi}{80}=\frac{\pi}{40}$$

To find the vertical shift D:

$$D=30+6=36\ feet$$

Write the equation for h(t):

$$h(t)=-30\cos\left(\frac{\pi}{40}t\right)+36$$

The maximum height is the vertical shift plus the radius:

$$h_{max}=36+30=66\ feet$$

Substitute $t=35$ into the equation:

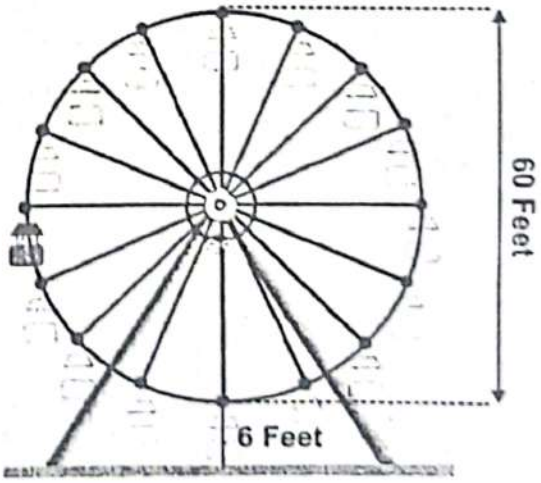
$$h(35)=-30\cos\left(\frac{\pi}{40}(35)\right)+36$$

$$h(35)=-30\cos\left(\frac{7\pi}{8}(35)\right)+36$$

$$h(35)\approx-30(-0.9239)+36$$

$$h(35)\approx27.717+36$$

$$h(35)\approx63.717\ feet$$



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(a) **Model an equation that represent the height h(t) of rider on the Ferris wheel at any given time t.**

The equation for the height of the rider is

$$h(t) = -30\cos\left(\frac{\pi}{40}t\right) + 36$$

(b) **Find the maximum height of a rider**

The maximum height of the rider is 66 feet.

(c) **Find the height of the rider from the ground after 35 seconds.**

The height of the rider after 35 seconds is approximately 63.717 feet

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Q6. **A child is playing on a swing in playground. The height h(t) of the swing seat above the ground (in meters) at time t (in seconds) is modeled by the function.**

$$h(t) = 1.5 + 1.2\sin(3\pi t)$$

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a) **What is the maximum height reached by the swing seat?**

Sol: The height function is given by $h(t) = 1.5 + 1.2\sin(3\pi t)$. The sine function, $\sin(3\pi t)$, oscillates between -1 and 1. To find the maximum height, we need to find the maximum value of $h(t)$. This occurs when $\sin(3\pi t)$ reaches its maximum value, which is 1.

So, the maximum height is:

$$h_{max} = 1.5 + 1.2(1) = 1.5 + 1.2 = 2.7 \text{ meters.}$$

(b). **what is the minimum height reached by the swing seat?**

Sol: Similarly, the minimum height occurs when $\sin(3\pi t)$ reaches its minimum value, which is -1.

So, the minimum height is:

$$h_{min} = 1.5 + 1.2(-1) = 1.5 - 1.2 = 0.3 \text{ meters.}$$

(c). **How long does it take for the swing to complete one full back-and-forth motion (period)?**

Sol: The period of a sinusoidal function of the form $\sin(Bt)$ is given by $T = \frac{2\pi}{|B|}$. in our case, $B = 3\pi$.

Therefore, the period fo the swing is:

$$T = \frac{2\pi}{|3\pi|} = \frac{2\pi}{3\pi} = \frac{2}{3} \text{ seconds.}$$

This means it takes $\frac{2}{3}$ seconds for the swing to complete on full back-and-forth motion.

(d). **At what time (s) does the swing seat first reach a height of 2.12 meters?**

Sol: We wand to find the time(s) t when h(t)=2.12 meters. So we set up the equation

$$2.12 = 1.5 + 1.2\sin(3\pi t)$$

Now, let's solve for $\sin(3\pi t)$

$$2.12 - 1.5 = 1.2\sin(3\pi t)$$

$$0.62 = 1.2\sin(3\pi t)$$

$$\sin(3\pi t) = \frac{0.62}{1.2} = \frac{31}{60}$$

Now we need to find the values of t for which $\sin(3\pi t)$. first, let's find the principal value:

$$3\pi t = \arcsin\left(\frac{31}{60}\right)$$

$$t = \frac{1}{3\pi}\arcsin\left(\frac{31}{60}\right)$$

Using a calculator, $\arcsin\left(\frac{31}{60}\right) \approx 0.5435$ radians.

So, the first time the swing reaches 2.12 meters is:

$$t_1 = \frac{0.5435}{3\pi} \approx \frac{0.535}{9.4248} \approx 0.0577 \text{ Seconds.}$$

Therefore, the swing seat first reaches a height of approximately 0.577 seconds.

Q7. **A carnival ride consist of a vertical wheel with a diameter of 40 feet. The centre of the wheel is 28 feet above the ground. The wheel rotates at a constant speed and takes 120 seconds to make one complete revolution. Model an equation that describes the height h(t) of a rider on the wheel as a function of time t. how high is the rider from the ground after 90 seconds? At what times will the rider be 36 feet above the ground?**

Sol: let's model the equation: $h(t) = a\cos(bt) + d$

Where:

(a) A=amplitude=radius=20 feet (half of diameter)

D=vertical shift =28 feet (center of wheel above ground)

Period = 120 seconds, so $b = 2\pi / 120 = \pi / 60$

$$h(t) = -20\cos(\pi t / 60) + 28$$

Note the negative sign, as the cosine function starts at its maximum value, but we want it to start at the minimum value (lowest point).

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(b) **Height after 90 seconds:**

$$h(90) = -20\cos(\pi(90)/60) + 28$$

$$= -20\cos(3\pi/2) + 28 = -20(0) + 28 = 28\ feet$$

(c) **Time when rider is 36 feet above ground:**

$$36 = -20\cos(\pi t/60) + 28$$

$$8 = -20\cos(\pi t/60)$$

$$\cos(\pi t/60) = -8/20 = \frac{-2}{5}$$

Cos function is negative in II and III quad So,

$$\frac{\pi}{60}t = \pi - \cos^{-1}\left(\frac{2}{5}\right)$$

$$\Rightarrow t = \frac{60}{\pi}\left(\pi - \cos^{-1}\frac{2}{5}\right) = 37.865 \text{ in II quad}$$

$$\frac{\pi}{60}t = \pi + \cos^{-1}\left(\frac{2}{5}\right)$$

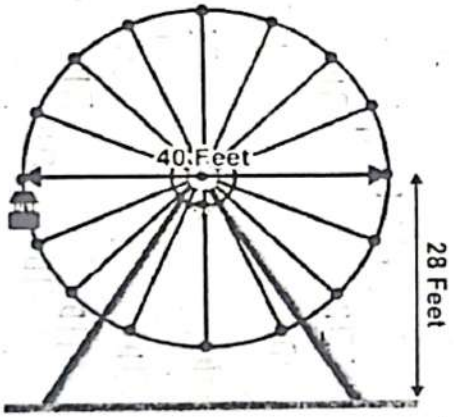
$$\Rightarrow t = \frac{60}{\pi}\left(\pi + \cos^{-1}\frac{2}{5}\right) = 82.145 \text{ in III quad}$$

The equation is $h(t) = -20\cos(\pi t/60) + 28.$

Height after 90 seconds: 28 feet

Times when rider is 36 feet above ground:

Approximately 37.86 seconds and 82.14 seconds



Q8. **Suppose the temperature T in degrees Fahrenheit of Lahore city in a month of December throughout the day can be modeled by the equation:**

$$T = 64 + 8\sin\left(\frac{\pi}{12}(t-8)\right),$$ **where t is the time in hours. The temperature oscillates 8 degrees above and below and average temperature of 64 degrees.**

(a) Find the temperature at $t = 9$ hours?

(b) At what time the temperature will be maximum?

(c) Calculate the maximum temperature.

Sol: (a) **Temperature at $t = 9$ hours:**

Start with the equation:

$$T = 64 + 8\sin\left(\frac{\pi}{12}(t-8)\right)$$

Substitute $t = 9$:

$$T = 64 + 8\sin\left(\frac{\pi}{12}(9-8)\right)$$

$$T = 64 + 8\sin\left(\frac{\pi}{12}(1)\right)$$

$$T = 64 + 8\sin\left(\frac{\pi}{12}\right)$$

$$\Rightarrow \sin\left(\frac{\pi}{12}\right) \approx 0.2588$$

Multiply by 8:

$$8 \times 0.2588 \approx 2.0704$$

$$T \approx 64 + 2.0704 \approx 66.07\,F^\circ$$

The temperature at $t = 9$ hours is approximately 66.07 degrees Fahrenheit.

(b) **Time at which the temperature will be maximum**

$\sin\left(\frac{\pi}{12}(t-8)\right)$ reaches its maximum value which is 1.

Set the sine term equal to 1.

$$\sin\left(\frac{\pi}{12}(t-8)\right) = 1$$

Find the angle for which sine is 1.

The first such angle is $\frac{\pi}{12}$

$$\frac{\pi}{12}(t-8) = \frac{\pi}{2}$$

Solve for $(t-8)$:

$$t - 8 = \frac{\cancel{\pi}}{2} \times \frac{12}{\cancel{\pi}} = 6$$

Solve for t:

$t = 6 + 8 = 14h$ or 2pm

The temperature will be maximum at $t = 14$ hours

(c) **Maximum temperature:**

$$T = 64 + 8\sin\left(\frac{\pi}{12}(t-8)\right)$$

Here a $a = 64,\ b = 8$

Max temp $= M = a + |b|$

$= 64 + 8$

$= 72^\circ$

The maximum temperature is 72 degrees Fahrenheit

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Q9. Suppose the population of a coastal city follows a sinusoidal pattern due to seasonal migration. The population of the city over the course of a year can be modeled by the equation:

$P(t) = 70000 + 10000\left(\frac{\pi}{6}t - \frac{\pi}{2}\right)$, $P(t)$ is the population at time t (t is the time in months, with t = 0 corresponding to January 1st). Where t denoted the months in a year.

(a) Find the population of a city at $t = 7$ months

(b) Find the maximum population

Sol: (a) Calculate the population at $t = 7$

Substitute $t = 7$ into the population model:

$$P(7) = 70000 + 10000\cos\left(\frac{\pi}{6}(7) - \frac{\pi}{2}\right)$$

$$P(7) = 70000 + 10000\cos\left(\frac{7\pi}{6} - \frac{3\pi}{6}\right)$$

$$P(7) = 70000 + 10000\cos\left(\frac{4\pi}{6}\right)$$

$$P(7) = 70000 + 10000\cos\left(\frac{2\pi}{3}\right)$$

$$P(7) = 70000 + 10000\left(-\frac{1}{2}\right)$$

$$P(7) = 70000 - 5000$$

$$P(7) = 65000$$

(b) Calculate the maximum population

$$P(t) = 70000 + 10000\cos\left(\frac{\pi}{6}t - \frac{\pi}{2}\right)$$

Here a $a = 70000,\ b = 10000$

Max temp $= M = a + |b|$

$P_{max} = 70000 + 10000$

$P_{max} = 80000$

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